

$\widetilde{\text{Optimal}}$ Collective Tree Spanners

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Abstract

Graph distance spanners are fundamental algorithmic tools which preserve graph distances up to some error or distortion with few edges. A collective tree spanner is a collection of spanning trees such that, for any pair of vertices, some tree contains a low-stretch path between them. We identify a simple construction for a collective tree $(2k - 1)$ -spanner with $\widetilde{O}(kn^{1/k})$ spanning trees, which is optimal for fixed k up to polylogarithmic factors assuming the Erdős girth conjecture.

1 Introduction

A *collective tree α -spanner* (cf. spanning tree cover) of a graph $G = (V, E)$ is a collection $\mathcal{T}$ of spanning trees of G such that for any pair $u, v \in V$ of vertices, there is some tree $T \in \mathcal{T}$ for which,

$$d_T(u, v) \leq \alpha \cdot d_G(u, v)$$

Extending a long line of work on *additive* collective tree spanners [CDKY05, DYL06, DYC06, DL07, DY10, DCKX12] and tree covers [TZ05, MN07, BFN22], the problem of constructing multiplicative collective tree spanners has recently been considered and has applications in routing [GKR05, CCL⁺25], differential privacy [CZ25], and online algorithms [GKLO16].

Problem 1.1 (cf. [ACE⁺20]). Does there exist a spanner that is a union of a sublinear in n number of trees, such that every pair of vertices has a low stretch path in one of these trees?

Abraham et al. [ACE⁺20] first answered this problem using a Ramsey-type theorem for graphs, constructing an $O(k \log \log n)$ spanner with $kn^{1/k}$ spanning trees. Up until now, their construction was the best-known for general graphs. We note that the spanner of [ACE⁺20] is actually a *tree-padding spanner*, which has the additional property that each vertex $v \in V(G)$ has a single tree T_v which contains a low-stretch path from v to all other vertices. By relaxing this constraint, we are able to achieve a nearly optimal size/stretch tradeoff.

Theorem 1.2. *Let G be a connected undirected n -vertex graph with positive edge weights, and let $k \geq 2$ be an integer. There is a randomized construction which, with probability at least $1 - n^{-4}$, outputs a collective tree $(2k - 1)$ -spanner with*

$$O\left(kn^{1/k}(\log k)^{1-1/k} \log n\right)$$

spanning trees. The construction runs in $O_k(n^{1/k} \log n (m + n \log n))$ time.

Proof overview. Our spanner construction is a modification of the well-known distance oracle of Thorup and Zwick [TZ05]. Their oracle constructs vertex sets called *clusters*, with the property that for every pair u, v of vertices, there is a cluster centered at w containing both u and v , and

$$d_G(u, w) + d_G(w, v) \leq (2k - 1)d_G(u, v).$$

One shortest-path tree per cluster would suffice, but the union of the overlapping trees need not be a forest; it may contain cycles. We instead color the centers and assign every vertex to its nearest center of each color. The resulting *Voronoi cells* are disjoint, and their shortest-path trees form a forest that can be completed to a spanning tree of G .

We must ensure that w wins the competitions of being the closest center of some fixed color to both u and v . We show the number of competitors with w has a geometric tail and is therefore small with constant probability. With $\Theta(n^{1/k}(\log k)^{1-1/k})$ colors, w then wins the competition at both endpoints with constant probability. Thus, $\Theta(\log n)$ repetitions cover all pairs with high probability.

Relation to tree covers. A collective tree spanner is a strengthening of a *tree cover*. For graphs, a tree cover is exactly a collective tree spanner with the relaxation that the collection of trees need not individually span the entire graph, and the edges in the cover need not be present in the original graph. A Ramsey tree cover additionally gives every vertex one home tree that preserves all pairs incident to it [MN07, ACE⁺20, BFN22]. Gupta, Kumar, and Rastogi famously introduced the *spanning tree cover* model for multiprotocol label switching [GKR05]. Additive collective tree spanners were later developed for several graph classes [DYL06, CDKY05, DY10, DCKX12]. More recently, Chang et al. constructed light $(1 + \varepsilon)$ -stretch spanning tree covers for doubling graphs and used them to obtain compact routing schemes and path-reporting distance oracles [CCL⁺25]. Elkin and Shabat subsequently constructed $(1 + \varepsilon)$ -stretch spanning tree covers for p -path-separable graphs, together with larger-stretch tradeoffs using fewer trees [ES26]. Known applications of multiplicative collective tree spanners include routing [GKR05, CCL⁺25], differentially private shortest-path queries [CZ25], and advice complexity for the k -server problem [GKLO16].

Lower bounds. Under the Erdős girth conjecture, there are unit-weight n -vertex graphs of girth greater than $2k$ with $\Omega(n^{1+1/k})$ edges (cf. [ABS⁺20]). Every $(2k - 1)$ -spanner of such a graph must keep every edge: after deleting uv , any replacement $u \rightarrow v$ path has at least $2k$ edges. The union of t trees in a collective tree spanner is itself a spanner and has at most $t(n - 1)$ distinct edges, forcing $t = \Omega(n^{1/k})$. Theorem 1.2 matches this lower bound up to logarithmic factors when k is constant. In the metric model, for every fixed t , Chen, Tan, and Xu exhibit an n -point metric for which every size- t non-Steiner tree cover has distortion $\Omega_t(n^{1/2^{t-1}})$ [CTX26].

2 Spanner construction

Let $G = (V, E)$ be connected, let $n = |V|$, and give every edge a positive weight. Write d_G for shortest-path distance. Fix a bijection $\mu : V \rightarrow \{1, 2, \dots, |V|\}$. For a vertex x , write

$$w \prec_x z \iff (d_G(x, w), \mu(w)) < (d_G(x, z), \mu(z))$$

in lexicographic order. For nonempty $A \subseteq V$, let $p_A(x)$ be the $\prec_x$ -minimum element of A . When $A = \emptyset$, define $p_A(x) = \perp$, where every vertex precedes $\perp$. We adopt this ordering convention to handle distance-ties between pairs of vertices.

Thorup and Zwick distance oracle

The simple distance oracle of Thorup and Zwick [TZ05] works by randomly sampling a *hierarchy* of vertex sets

$$A_0 = V \supseteq A_1 \supseteq \dots \supseteq A_{k-1} \supseteq A_k = \emptyset, \quad P_i = A_i \setminus A_{i+1}$$

and defining for each $w \in P_i$ the *cluster*,

$$C_i(w) = \{x \in V : w \prec_x p_{A_{i+1}}(x)\}$$

They then prove the somewhat surprising result that for each pair of vertices, there is some cluster containing a low-stretch path between them.

Lemma 2.1 (Thorup and Zwick [TZ05]). *For every $u, v \in V$, there are an index $i \in \{0, \dots, k-1\}$ and a center $w \in P_i$ such that $u, v \in C_i(w)$ and*

$$d_G(u, w) + d_G(w, v) \leq (2k-1)d_G(u, v).$$

Proof. Set $\Delta := d_G(u, v)$. We alternately search from the two endpoints. At stage i , consider the ordered pair (x_i, y_i) , beginning with $(x_0, y_0) = (u, v)$, and let

$$w_i := p_{A_i}(x_i).$$

Stop at the first stage for which

$$w_i \prec_{y_i} p_{A_{i+1}}(y_i). \tag{1}$$

If the test fails, swap the endpoints for the next stage: $(x_{i+1}, y_{i+1}) \leftarrow (y_i, x_i)$. If $A_{i+1} = \emptyset$, then the right-hand side of (1) is $\perp$ and the test succeeds. Thus, a failed stage has $A_{i+1} \neq \emptyset$. In particular, the process must stop by stage $k-1$, at which stage $p_{A_k}(y_{k-1}) = \perp$, which w_{k-1} precedes.

We first bound how far the current center can be from the endpoint that chose it. We claim that

$$d_G(x_i, w_i) \leq i\Delta. \tag{2}$$

For $i = 0$, we have $w_0 = p_V(x_0) = x_0$, so the claim is immediate. Suppose stage i fails. Then $p_{A_{i+1}}(y_i) \preceq_{y_i} w_i$. Because $x_{i+1} = y_i$ and $w_{i+1} = p_{A_{i+1}}(x_{i+1})$, the triangle inequality gives

$$\begin{aligned} d_G(x_{i+1}, w_{i+1}) &= d_G(y_i, p_{A_{i+1}}(y_i)) \\ &\leq d_G(y_i, w_i) \\ &\leq d_G(y_i, x_i) + d_G(x_i, w_i) \\ &\leq (i+1)\Delta. \end{aligned}$$

This proves (2) by induction.

Now let i be the stopping stage. The center w_i cannot belong to A_{i+1} : if it did, the first A_{i+1} -vertex under $\prec_{y_i}$ would satisfy $p_{A_{i+1}}(y_i) \preceq_{y_i} w_i$, contradicting (1). Therefore $w_i \in A_i \setminus A_{i+1} = P_i$. Moreover, $A_{i+1} \subseteq A_i$, while $w_i = p_{A_i}(x_i)$ and $w_i \notin A_{i+1}$. Thus every A_{i+1} -vertex follows w_i under $\prec_{x_i}$, which implies $w_i \prec_{x_i} p_{A_{i+1}}(x_i)$, so $x_i \in C_i(w_i)$. The stopping condition gives $y_i \in C_i(w_i)$ as well.

Finally, $\{x_i, y_i\} = \{u, v\}$, and hence

$$\begin{aligned} d_G(u, w_i) + d_G(w_i, v) &= d_G(x_i, w_i) + d_G(w_i, y_i) \\ &\leq 2d_G(x_i, w_i) + \Delta \\ &\leq (2i+1)\Delta \leq (2k-1)\Delta, \end{aligned}$$

where the first inequality follows from the triangle inequality. $\square$

Packing clusters into spanning trees

First, we argue that partitioning V by color classes R constructs Voronoi cells, each of which contains a shortest-path tree rooted at the center.

Lemma 2.2. *Let $R \subseteq V$ be nonempty. Assign each $x \in V$ to its $\prec_x$ -minimum root $r(x) \in R$. Every cell*

$$D_R(w) = \{x : r(x) = w\}$$

contains a shortest $w \rightarrow x$ path for each $x \in D_R(w)$. Consequently the cells admit vertex-disjoint shortest-path trees whose union is a forest spanning V . This forest extends to a spanning tree of G .

Proof. Let $x \in D_R(w)$ with $x \neq w$, and let y be the next vertex after x on a shortest x -to- w path. If a root z preceded w from y , then

$$d_G(x, z) \leq d_G(x, y) + d_G(y, z) \leq d_G(x, y) + d_G(y, w) = d_G(x, w).$$

A strict inequality would make z closer than w from x . Likewise, distance equality and a smaller μ rank would make z win the tie at x . Both contradict $x \in D_R(w)$. Thus $y \in D_R(w)$. Iterating this argument gives shortest $w \rightarrow x$ path contained in the cell. Thus, the cell contains a shortest-path tree rooted at w spanning $D_R(w)$. The union of these spanning trees is a spanning forest of G . Add edges arbitrarily between components until it becomes a spanning tree of G . $\square$

Set¹

$$a = \ln(32k), \quad p = \left(\frac{a}{n}\right)^{1/k}, \quad \tau = \left\lceil \frac{a}{p} \right\rceil, \quad q = 32\tau. \quad (3)$$

One trial of our construction performs the following steps.

- (i) Set $A_0 = V$. For each $i = 0, \dots, k-2$, independently retain every vertex of A_i in A_{i+1} with probability p . Set $A_k = \emptyset$ and $P_i = A_i \setminus A_{i+1}$.
- (ii) Assign each center in each P_i a color chosen independently and uniformly from $\{1, \dots, q\}$.
- (iii) For every level i and color c , let

$$R_{i,c} := \{w \in P_i : w \text{ has color } c\}.$$

If $R_{i,c}$ is nonempty, form Voronoi forest from Lemma 2.2 and extend that forest to a spanning tree of G .

Thus a trial creates at most kq spanning trees. For $x \in V$, define the level- i bunch

$$S_i(x) := \{w \in P_i : x \in C_i(w)\}.$$

We argue that the size of these bunches has a geometric tail.

Lemma 2.3. *For $0 \leq i \leq k-2$, every $x \in V$, and every integer $s \geq 0$,*

$$\Pr[|S_i(x)| \geq s] \leq (1-p)^s \leq e^{-ps}.$$

and at level $k-1$,

$$S_{k-1}(x) = A_{k-1}, \quad \mathbb{E}|A_{k-1}| = \frac{a}{p} \leq \tau.$$

¹We may assume that $n > a$ such that $p \leq 1$. If $n \leq a$, taking the shortest-path tree rooted at every vertex satisfies the conclusion of the main theorem.

Proof. Fix $i \leq k - 2$ and condition on the set A_i . List its vertices in their $\prec_x$ -order as

$$v_1 \prec_x v_2 \prec_x \cdots \prec_x v_r.$$

Let j be the first index for which v_j is retained in A_{i+1} , with $j = r + 1$ if no vertex is retained. Then $p_{A_{i+1}}(x) = v_j$ when $j \leq r$, and by the definition of a cluster,

$$S_i(x) = \{v_1, \dots, v_{j-1}\}.$$

If $s > r$, the event $|S_i(x)| \geq s$ is impossible. Otherwise, it requires the first s vertices in the list to be rejected. Conditional on A_i , those choices are independent, so this event has probability at most $(1 - p)^s$. The standard inequality $1 - p \leq e^{-p}$ gives the second bound.

At level $k - 1$, we have $P_{k-1} = A_{k-1}$ and $A_k = \emptyset$, so every center precedes $p_{A_k}(x) = \perp$. Thus $S_{k-1}(x) = A_{k-1}$. Each vertex of V survives the first $k - 1$ sampling steps with probability p^{k-1} . Linearity of expectation gives,

$$\mathbb{E}|A_{k-1}| = np^{k-1} = \frac{np^k}{p} = \frac{a}{p} \leq \tau.$$

□

Since the bunches $S_i(x)$ are *small*, there are few competitors that can precede w at either endpoint u or v . Thus, w wins the competition against other centers with constant probability.

Lemma 2.4. *For every fixed pair $u, v \in V$, with probability greater than $1/2$ some tree T generated in one trial satisfies $d_T(u, v) \leq (2k - 1)d_G(u, v)$.*

Proof. Call the hierarchy good for u, v if

$$\begin{aligned} \max\{|S_i(u)|, |S_i(v)|\} &\leq \tau & (0 \leq i \leq k - 2), \\ |A_{k-1}| &\leq 4\tau. \end{aligned}$$

Because $p\tau \geq a$, Lemma 2.3 implies, for every level except $i = k - 1$ and either endpoint $x \in \{u, v\}$,

$$\Pr[|S_i(x)| > \tau] \leq e^{-p(\tau+1)} \leq e^{-a} = \frac{1}{32k}.$$

At the final level $i = k - 1$, Markov's inequality gives

$$\Pr[|A_{k-1}| > 4\tau] \leq \frac{\mathbb{E}|A_{k-1}|}{4\tau} \leq \frac{1}{4}.$$

A union bound over the $2(k - 1)$ bunches when $i \leq k - 2$ therefore yields

$$\begin{aligned} \Pr[\text{the hierarchy is not good}] &\leq \frac{2(k - 1)}{32k} + \frac{1}{4} \\ &\leq \frac{5}{16}. \end{aligned}$$

Hence the hierarchy is good with probability at least $11/16$.

Condition on a good hierarchy, and fix a low-stretch center $w \in P_i$ from Lemma 2.1. Define the centers that could take an endpoint away from w by

$$B := \{z \in P_i \setminus \{w\} : z \prec_u w \text{ or } z \prec_v w\}.$$

If $z \prec_u w$, then $u \in C_i(w)$ gives

$$z \prec_u w \prec_u p_{A_{i+1}}(u),$$

and hence $u \in C_i(z)$ and $z \in S_i(u)$. The same argument applies at v . Therefore $B \subseteq S_i(u) \cup S_i(v)$. If $i \leq k-2$, goodness of the hierarchy gives $|B| \leq 2\tau$. If $i = k-1$, then $B \subseteq A_{k-1}$ and $|B| \leq 4\tau$. Thus, uniformly over all levels, $|B| \leq 4\tau$.

Now condition also on the color of w . Each center in B independently receives that same color with probability $1/q$. By a union bound,

$$\Pr[\text{no center in } B \text{ has the color of } w] \geq 1 - \frac{|B|}{q} \geq 1 - \frac{4\tau}{q} \geq \frac{7}{8}.$$

On this event, w is the first root of its color from both u and v . Thus $u, v \in D_{R_i, c}(w)$, where c is the color of w . Let T be the spanning tree constructed from this color class. Lemma 2.2 ensures that T preserves the two distances to w , so

$$\begin{aligned} d_T(u, v) &\leq d_T(u, w) + d_T(w, v) \\ &= d_G(u, w) + d_G(w, v) \\ &\leq (2k-1)d_G(u, v), \end{aligned}$$

where the last inequality is Lemma 2.1. Since the coloring is independent of the hierarchy, the total success probability is at least

$$\frac{11}{16} \cdot \frac{7}{8} > \frac{1}{2}.$$

□

Now, the main theorem follows by boosting the success probability of Lemma 2.4 via multiple independent trials.

Proof of Theorem 1.2. Run

$$L := 6\lceil \log_2 n \rceil$$

independent trials, and return the collection of all trees they produce. By Lemma 2.4, a fixed vertex pair fails in every trial with probability less than

$$2^{-L} \leq n^{-6}.$$

By a union bound over all $\binom{n}{2}$ vertex pairs, the probability that any pair is uncovered is less than n^{-4} .

It remains to count the number of trees. Since $p^k = a/n$,

$$\begin{aligned} \frac{a}{p} &= n^{1/k} a^{1-1/k} \\ &= n^{1/k} (\ln(32k))^{1-1/k}. \end{aligned}$$

Each trial creates at most $kq = O(k\tau) = O(ka/p)$ trees. After boosting $L = \Theta(\log n)$ times, the total number of trees is,

$$O\left(kn^{1/k} (\log k)^{1-1/k} \log n\right).$$

Finally, we bound the running time. Let $m = |E|$. Sampling the hierarchy and assigning the colors takes $O(kn)$ time per trial. For each nonempty set $R_{i,c}$, its Voronoi forest can be constructed by one execution of multi-source Dijkstra's algorithm. With the standard implementation, this takes

$O(m + n \log n)$ time. Since each trial has at most kq nonempty color classes, the total running time over all L trials is

$$O(L(kn + kq(m + n \log n))) = O\left(kn^{1/k}(\log k)^{1-1/k} \log n (m + n \log n)\right).$$

□

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